

Macroeconomics I — MPE Insper

Mock Final Exam 3 (conceptual) — Simulado 3 — 2026, 3rd term

Instructions. Four blocks of 25 points (100 in total). Duration: 3 hours, closed book, non-programmable calculator allowed. Each block has one True/False question (two items, 5 points each) and one discursive question (items a, b, c, 5 points each).

True/False with justification. Mark T or F and justify. The verdict is worth **1 point** and the justification **4 points**. A verdict with no justification earns at most 1 point; a justification whose words contradict its own verdict or its own algebra earns 0. A good justification names the mechanism in one causal chain and, when the statement is false, says what is true instead.

Discursive items ask you to *explain, draw and interpret, sign and say which effect dominates*, or say *what an observer with ordinary data would wrongly conclude*. Each item ends with the list of objects your answer must contain: tick them off before moving on. Computation is asked for only where a short derivation is the explanation. Sign every derivative before interpreting it; box final answers; label axes, curves and the direction of every shift. Notation: Kurlat (2020) in Blocks I–III, Benigno (2015) in Block IV. All numbers are illustrative.

Solution. There is no answer key in this PDF. The worked solution, with every step, the figures and the grading rubric, is the set of linked notes in `Map/simulado-03/`, starting at `00-index.md`. Sit the exam timed first, then read.

Block I — Measurement and national accounts

(Lecture 1 • 25 points)

Question 1. (10 points) True or False? Justify.

a) (5 points) A small economy produces only streaming subscriptions and cinema tickets (prices in reais, quantities in millions):

	p_{stream}	q_{stream}	p_{cinema}	q_{cinema}
Year 1	40	10	20	30
Year 2	20	30	25	25

Statement: “A statistician who values both years’ quantities at year-2 prices will report faster real growth than one who uses year-1 prices, because year-2 prices are more up to date.” **T** ☐ **F** ☐

Justification:

b) (5 points) *Statement:* “In a year in which the only price that changes is the price of imported televisions, which rises by 30%, the GDP deflator rises by more than the consumer price index (IPCA).” **T** ☐ **F** ☐

Justification:

Question 2. (15 points) Brazil and Paraguay: which number answers which question? Over the last decade (illustrative figures), Paraguay's real GDP grew 3.5% a year and Brazil's 2.5% a year; Paraguay's population grew 2.0% a year and Brazil's 0.5%. Today GDP per capita converted at market exchange rates is USD 9,000 in Brazil and USD 6,000 in Paraguay. The price level relative to the United States (the ratio of the PPP exchange rate to the market rate) is 0.55 in Brazil and 0.40 in Paraguay. Employment is 45% of the population in Brazil and 50% in Paraguay.

a) (5 points) A newspaper headline reads: "Paraguay grows 40% faster than Brazil, so Paraguayans' living standards are catching up with Brazilians." Compute the growth rate of GDP per capita in each country (exactly and by the log approximation). Does the headline's conclusion follow? Explain why aggregate growth and per capita growth answer different questions.

Answer every object: g_y for Paraguay; g_y for Brazil; verdict on the headline; the question each growth rate answers.

Answer:

b) (5 points) Compute GDP per capita at PPP in both countries, and the Brazil/Paraguay ratio at market rates and at PPP. Explain, naming the mechanism, why the ratio shrinks, and say what an observer who uses market exchange rates would wrongly conclude about the gap in living standards.

Answer every object: two PPP values; two ratios; the mechanism (which goods, which prices, why); the wrong conclusion and its direction.

Answer:

c) (5 points) A consultant ranks the two countries' *labour productivity* by GDP per capita at PPP. Compute GDP per worker at PPP. Say which of three measures (aggregate GDP at market rates; GDP per capita at PPP; GDP per worker at PPP) answers each of: (i) the size of the market an exporter selling in dollars faces; (ii) average material living standards; (iii) labour productivity. Explain why per capita and per worker give different ratios here.

Answer every object: two per-worker values and their ratio; one measure for each of (i), (ii), (iii); the reason the two ratios differ.

Answer:

Block II — Long-run growth and the Solow model

(Lectures 2–3 • 25 points)

Question 3. (10 points) True or False? Justify. Solow model, variables per worker, no technological progress.

a) (5 points) *Statement:* “The Solow model predicts that poorer countries grow faster than richer ones. So a poor country that has grown more slowly than a rich one for decades is evidence against the model.” **T** ☐ **F** ☐

Justification:

b) (5 points) Take $\alpha = 1/3$, $s = 0.2$, $\delta = 0.05$. *Statement:* “A permanent fall in the population growth rate from 2% to 1% a year lowers, in the long run, both the growth rate of aggregate GDP and the growth rate of GDP per capita.” **T** ☐ **F** ☐

Justification:

Question 4. (15 points) Tax paperwork read as productivity. Brazilian firms spend an unusually large number of hours complying with tax rules. Model it as follows. Output is $Y = AK^\alpha L_P^{1-\alpha}$, where L_P is production labour. For each production worker, a firm must also employ φ compliance workers, who produce nothing and are paid the same wage w . Official statistics record total employment $L = (1 + \varphi)L_P$. Factor markets are competitive and the price of output is 1. Take $\alpha = 1/3$ and $\varphi = 0.25$.

a) (5 points) Write Y as a function of A , K , L and φ . An observer has accurate data on Y , K , L and on factor incomes, but cannot tell compliance workers from production workers. Which labour share and which productivity $\hat{A}$ does he measure? Is the labour share distorted? Where does the distortion show up?

Answer every object: $Y(A, K, L, \varphi)$; measured labour share, with the firm's first-order condition; $\hat{A}/A$ (number); share of the workforce in compliance; where the wedge sits.

Answer:

b) (5 points) A tax simplification lowers φ from 0.25 to 0.10 over five years. True A does not change. Write the growth-accounting decomposition of g_Y , solve for the Solow residual and compute its annual value. What does the statistical agency report, and why is this consistent with the reading of the residual as a measure that also captures policies that change how well resources are allocated? What would it report if compliance rules got *heavier*?

Answer every object: decomposition; residual in terms of φ ; its annual value; what is reported and why; the opposite case.

Answer:

c) (5 points) Now let the economy follow the Solow model with s , n and δ constant, per worker of *measured* employment. Draw the effect of the reform in two panels: the mechanism (Solow diagram) and the time path of y . By how much does steady-state y^* rise? Sign the growth rate of y on impact, during the transition and in the long run, and explain why the total rise in y exceeds the jump in measured TFP.

Answer every object: two-panel diagram; $y^{*'} / y^*$; growth of y in three horizons; the source of the extra gain.

Answer:

Block III — Microfoundations: consumption, labour and general equilibrium (Lectures 4–6 • 25 points)

Question 5. (10 points) True or False? Justify.

a) (5 points) A household has one unit of time, utility $\ln c + b \ln l$ over consumption and leisure, earns the after-tax wage $(1 - \tau)w$ per hour worked and receives a lump-sum transfer $T \geq 0$. *Statement:* “If the government returns nothing to households ($T = 0$), a rise in the labour-income tax τ reduces hours worked, because it makes leisure cheaper.” **T** ☐ **F** ☐

Justification:

b) (5 points) Matches are $m = \mu v^\alpha u^{1-\alpha}$ (in rates) and a fraction s of employed workers loses their job each period. *Statement:* “A fall in the unemployment rate together with a rise in the vacancy rate, with matching efficiency μ and the separation rate s unchanged, is an outward shift of the Beveridge curve: the market now has more vacancies per unemployed worker.” **T** ☐ **F** ☐

Justification:

Question 6. (15 points) Interest rates, a tax rebate and a tax on interest income. Households live two periods, have utility $u(c_1) + \beta u(c_2)$ with CRRA u (coefficient σ ; log when $\sigma = 1$), and borrow and lend at r unless a limit is stated.

a) (5 points) A saver has $y_1 = 100$, $y_2 = 0$, $\beta = 0.96$, $\sigma = 2$. The Copom raises the real rate from 4% to 6%. Sign $\partial c_1 / \partial r$, say which effect dominates and why, and compute c_1 at both rates. Then suppose *all* households are identical and capital and labour are fixed, so output is given in each period (frozen factors). Explain why aggregate saving must be zero, what then determines r , and compute r when $y_2 = y_1$ and when $y_2/y_1 = 1.03$.

Answer every object: sign; dominant effect in words that agree with the sign; c_1 at 4% and 6%; why aggregate saving is zero; what sets r ; the two values of r .

Answer:

b) (5 points) A young household has $y_1 = 40$, $y_2 = 84$, taxes $\tau_1 = 10$, $\tau_2 = 10.5$, log utility, $\beta = 1/1.05$ and $r = 5\%$. The government cuts τ_1 to 0 and raises τ_2 to 21. (i) With free borrowing, compute c_1 , c_2 and a before and after. (ii) Now impose $a \geq -15$. Compute c_1 , c_2 and a before and after. Does the timing of taxes matter in each case? Give two kinds of household for which (ii) is the relevant case.

Answer every object: c_1, c_2, a in all four cases; verdict on timing in (i) and in (ii), each with its reason; two examples.

Answer:

c) (5 points) A saver has $y_1 = 100$, $y_2 = 0$, log utility, $\beta = 0.96$, $r = 5\%$. The government taxes the return to saving: $c_2 = (1 + r - \tau_a)a$ with $\tau_a = 0.02$. Compare with a lump-sum tax in period 2 that raises the same revenue. Compute c_2/c_1 under each tax and the revenue. Which margin does each tax distort? Show that the household could afford the savings-tax plan under the lump-sum tax, and conclude which tax is better for the household.

Answer every object: revenue; both ratios c_2/c_1 ; the margin each tax distorts; the affordability argument; the welfare ranking.

Answer:

Block IV — Money, inflation and the New-Keynesian AS–AD model *(Lectures 7–9*

• 25 points)

Question 7. (10 points) True or False? Justify.

a) (5 points) *Statement:* “Last year M2 grew 10%, inflation was 6% and real GDP grew 3%. Since money grew faster than nominal GDP, these data refute the quantity equation $MV = PY$.” **T** ☐ **F** ☐

Justification:

b) (5 points) Money demand follows Baumol–Tobin, the real interest rate is 2% and real output does not grow. *Statement:* “Because money is neutral, a permanent rise in money growth from 4% to 10% a year has no effect on any real variable in the new steady state.” **T** ☐ **F** ☐

Justification:

Question 8. (15 points) The Copom, a drought and a payroll tax. Use Benigno (2015): AS is $p - p^e = \kappa(y - y_n)$ and AD is $y = \bar{y}_n - \sigma[i - (\bar{p} - p) - \rho]$, with $\bar{p} = p^e$. Take $\sigma = 0.5$, $\kappa = 1.133$, $\eta = 0.2$ and $\theta = 8$. Output and prices are in percent deviations from the initial natural level and from p^e .

a) (5 points) Money-market equilibrium is $M^S = p m^D(Y, i)$. Explain what this condition determines (i) when the central bank sets M and (ii) when it sets i . For an exogenous rise in M , say what adjusts with prices fixed and with prices flexible. Draw the two regimes. Why does Benigno's AS-AD model have no LM curve?

Answer every object: what is endogenous in (i) and in (ii); fixed-price and flexible-price answers; the two-panel diagram; the reason for no LM curve.

Answer:

b) (5 points) A drought temporarily lowers agricultural productivity so that y_n falls by 2% ($\bar{y}_n$ unchanged). The Copom keeps i unchanged. Which curve shifts, in which direction, and which curve does the economy move along? Compute dy , dp , $y - y_n$ and $y - y_e$. What change in i closes both gaps at once, and why can it?

Answer every object: the curve that shifts and its direction; the movement along the other; dy , dp ; both gaps with signs; the required di ; why one instrument suffices.

Answer:

c) (5 points) Instead, a temporary payroll-tax rise raises the mark-up by $d\mu = 4.4\%$, so that y_n falls by 2% while y_e does not move. The Copom minimises $L = (y - y_e)^2 + \frac{\theta}{\kappa}(p - p^e)^2$. Explain why the AS curve, and not the AD curve, is the constraint of this problem. Derive the optimal point and the change in i that delivers it, and explain why there is a trade-off here but not in (b).

Answer every object: why AS is the constraint; the targeting rule; $y - y_e$ and $p - p^e$ at the optimum; di ; the trade-off argument.

Answer: